Enriched Category Theory of Language

Enriched Category Theory and Distributional Measures of Language

Beyond Discrete Categories: Quantitative Grading

The categories introduced in **??**—with case roles as objects and grammatical relations as morphisms—capture the *qualitative* structure of case systems: which roles exist and how they connect. But linguistic data is fundamentally *quantitative*: some grammatical relations are more probable than others, some role assignments are stronger than others, and distributional similarity is a matter of degree.

To accommodate this quantitative dimension, we move from ordinary categories to **enriched categories**, where hom-sets are not just sets of morphisms but carry additional algebraic structure. The framework of Bradley et al. [-@fritz2021enriched] provides the key construction.

The $[0, 1]$-Enrichment of Case Categories

Definition

A category $\mathcal{C}$ enriched over the unit interval $([0, 1], \cdot, 1)$ assigns to every pair of objects A, B a *hom-value* $\mathcal{C}(A, B) \in [0, 1]$ satisfying:

$$\mathcal{C}(A, A) = 1 \quad \text{(Identity)} \tag{5.1}$$

$$\mathcal{C}(A, C) \geq \mathcal{C}(A, B) \cdot \mathcal{C}(B, C) \quad \text{(Composition)} \tag{5.2}$$

The identity axiom says that every expression is maximally related to itself. The composition inequality says that distributional relatedness composes sub-multiplicatively: if A is 80% related to B and B is 70% related to C, then A must be at least $0.8 \times 0.7 = 56\%$ related to C.

Interpreting Hom-Values in Linguistic Context

Bradley et al. [-@fritz2021enriched] interpret hom-values as *conditional probabilities* in a distributional model: $\mathcal{C}(A, B) = P(\text{context} \mid A \text{ and } B \text{ co-occur})$. In our case-theoretic application, we interpret them more broadly:

Categorical Magnitude as a Complexity Invariant

A key invariant of enriched categories is their **magnitude**—a numerical quantity that captures the "effective size" of the category, discounting for overlap between objects.

For an enriched category with n objects, let Z be the $n \times n$ matrix with $Z_{ij} = \mathcal{C}(i, j)$. The magnitude is:

$$|\mathcal{C}| = \sum_{i,j} (Z^{-1})_{ij} \tag{5.3}$$

when Z is invertible. Magnitude has deep connections to information theory:

- For a *discrete* category (no non-trivial relationships), $|\mathcal{C}| = n$ (the number of objects)
- For a highly connected category, $|\mathcal{C}| < n$ (objects are "redundant")
- Magnitude connects to the diversity measures studied in ecology (species diversity), graph theory (effective graph

Enriched Functors, Bradley's Language-as-Category Model, and the LLM Connection

Bradley's [-@bradley2024ipam; -@bradley2025tea] broader program treats natural language itself as an enriched category, where:

- **Objects** are expressions (words, phrases, sentences)
- **Hom-values** encode distributional co-occurrence probabilities
- **Composition** models transitivity of distributional relatedness

This "language as enriched category" perspective has profound implications for case theory:

1. **Case roles emerge from distributional structure**: Rather than being imposed a priori, case distinctions arise from clusters of high hom-values in the enriched language category. Nouns that frequently appear in agent contexts cluster together, forming the "nominative" region of the category.

2. **Alignment types correspond to enriched structure**: Different languages partition the enriched category differently, and these partitions correspond to the alignment types

Connection to Lawvere's Metric Spaces and Generalized Logic

The $[0,1]$-enrichment connects to a deep tradition in categorical algebra. Lawvere showed that metric spaces are categories enriched over $([0,\infty],+,0)$: the hom-value is the distance between points, the identity axiom says $d(x,x)=0$, and the composition inequality is the triangle inequality $d(x,z) \leq d(x,y)+d(y,z)$. Our $[0,1]$-enrichment is the multiplicative analogue: hom-values are *similarities* rather than distances, and the composition inequality is sub-multiplicative rather than sub-additive.

This Lawvere-style perspective unifies our case categories with the geometry of distributional semantics: case roles are points in a "similarity space," and morphisms between them are paths weighted by distributional proximity. The magnitude of this space then quantifies the "effective dimensionality" of the case system—how many independent relational distinctions the language makes.